
APIx

A quality-adjusted, high-frequency airfare price index for India — and the auditable infrastructure that produces it.

Pre-product engineering dossier. Prepared against MoSPI problem statement 26056, Data Informatics & Innovation Division.

Problem statement	Category	Document	Status
26056 · MoSPI DIID	Software · Smart Automation	Engineering dossier v2.1	Pre-build · gated

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Figures marked illustrative or indicative are placeholders. They must be replaced with measured values before external presentation.

The thesis

A fare appearing on a website is an observation. It is not automatically a price index observation. APIx is the infrastructure that decides, transparently and reproducibly, whether that observation is statistically comparable — and how much it should contribute to a measure of inflation.

India's revised Consumer Price Index, base 2024=100, went live on 12 February 2026. It adopts the COICOP 2018 framework, derives weights from HCES 2023–24, and — critically for this project — already collects airfare prices from online platforms rather than from ticketing counters.

The problem statement describes a transition MoSPI has already begun. APIx is not a proposal to start it. It is the measurement, engineering and validation infrastructure that makes it operate at production scale, at daily frequency, with quantified uncertainty and end-to-end auditability.

Who this is for

The primary user is a statistical analyst evaluating whether a series is adoptable: someone who will ask what the elementary aggregate is, where the weights came from, and what happens when a cell disappears. Every screen and every API field is designed to answer that person without a phone call. The secondary user is a monetary-policy economist who needs the series, its revision status and its uncertainty.

What this is not

Not a fare-prediction tool for travellers. Not a price-comparison service. Not a regulatory compliance instrument — APIx observes and reports, it does not adjudicate. Anything that would make the system more useful to a consumer and less legible to a statistician is out of scope by construction.

8.796

Weight of the Transport division in CPI 2024 (combined), up from 6.394 under CPI 2012 — a 38% increase in the division APIx serves.

Weekly

Current collection frequency for online prices under CPI 2024. Market prices are monthly. APIx targets daily observation on weekly-matched items.

78

Routes monitored monthly by DGCA's Tariff Monitoring Unit via airline websites, covering roughly 27% of domestic traffic.

Airfare is the weakest measurement point in the transport basket

Every other transport and communication item in CPI 2024 is priced from an administrative source with a single authoritative provider. Airfare is the exception. It is the only item in its neighbourhood collected from commercial, dynamically-priced consumer websites.

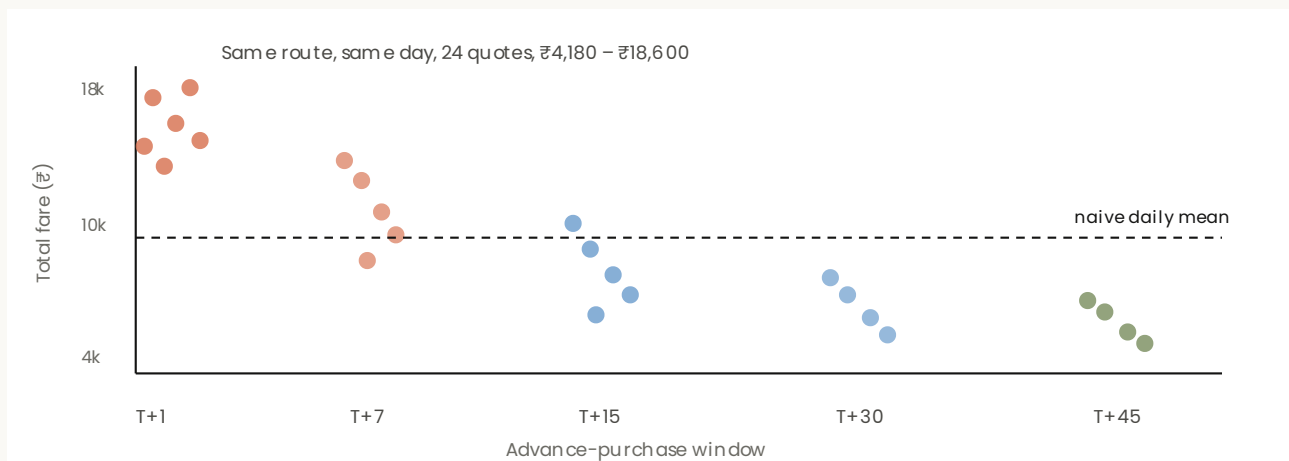
CPI 2024 item	Price source	Provider	Heterogeneity
Rail fares	Administrative data	Ministry of Railways	Low — published tariff
Fuel (petrol, diesel, LPG, CNG, PNG)	Administrative data	PPAC, MoPNG	Low — retail selling price
Telephone (mobile)	Plan prices + admin subscriber shares	Operators + TRAI	Low — fixed plans
Postage, money order	Administrative data	Department of Posts	None — regulated tariff
Airfare	Commercial online platforms	Airlines and aggregators	Extreme — dynamic pricing

This is not a criticism of the framework. It is a statement of where the measurement problem lives. An administrative artifact does exist: under a DGCA direction of 13 May 2025, scheduled domestic airlines publish tariff sheets conspicuously and furnish route-wise tariff categories to the regulator on the first of each month. But a declared fare category is not a transacted consumer price. The price a household actually pays is generated continuously by revenue-management systems responding to demand, and no administrative feed reports it.

Airfare is the CPI transport item where the collected price is market-facing and irreducibly heterogeneous, and where no administrative source reports the transacted consumer price. APIx is designed around that specific challenge.

Which of these is the price?

The problem statement observes that a single sector can vary by 200–400% within a day. That is true, and it is the crux of the engineering problem — but the variation is not all inflation. Much of it is *product mix*: quotes collected at different advance-purchase windows, in different fare families, on different departure times, with different baggage and refundability terms.



Illustrative. A single day's quote cloud on one sector. The dashed line is what a naive average reports. It moves when the composition of the cloud moves, even if no comparable product changed price.

The consequence

A daily-chained index built on naive averages accumulates composition effects. Over a window in which comparable fares return to their starting level, such an index does not return to its starting value. This is chain drift, and it is a bias, not noise. It does not average out with more data.

The solution: two estimators, one deterministic spine

APIx publishes two index series from the same canonical observations, computed by two parallel estimators that are never chained together.

APIx-L — the headline

Jevons short index at the elementary level, Young / Modified Laspeyres at higher levels. This is the aggregation architecture MoSPI itself uses in CPI 2024. Deterministic, formula-transparent, no machine learning anywhere in the code path.

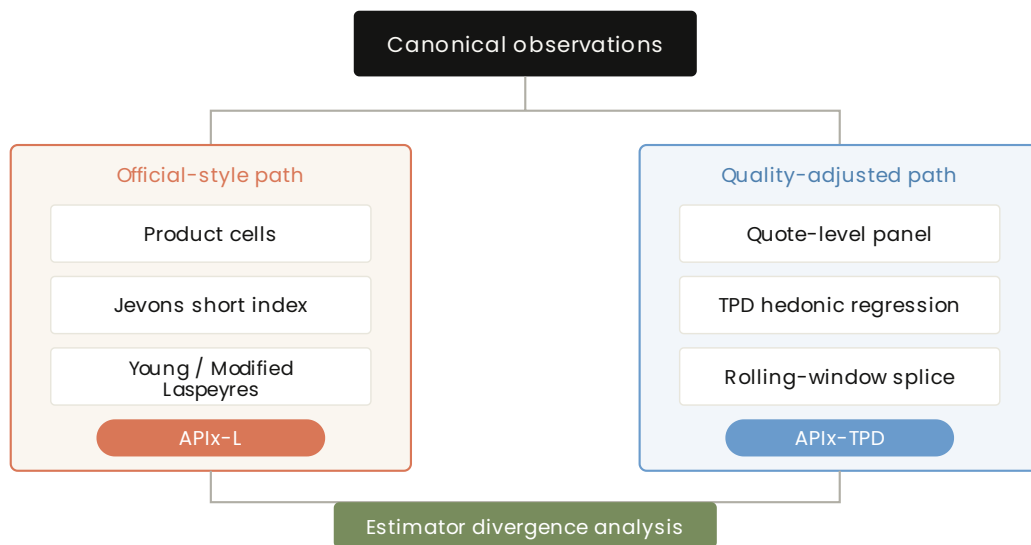
Purpose: a series a statistical office could adopt without changing its methodology.

APIx-TPD — the quality-adjusted estimate

A quote-level Time Product Dummy hedonic regression over a rolling window, spliced forward. Estimates price movement conditional on observed product characteristics.

Purpose: an independent quality-adjusted estimate of price movement under a specified hedonic model. It is another estimator under assumptions, not a detector of what is "really" happening.

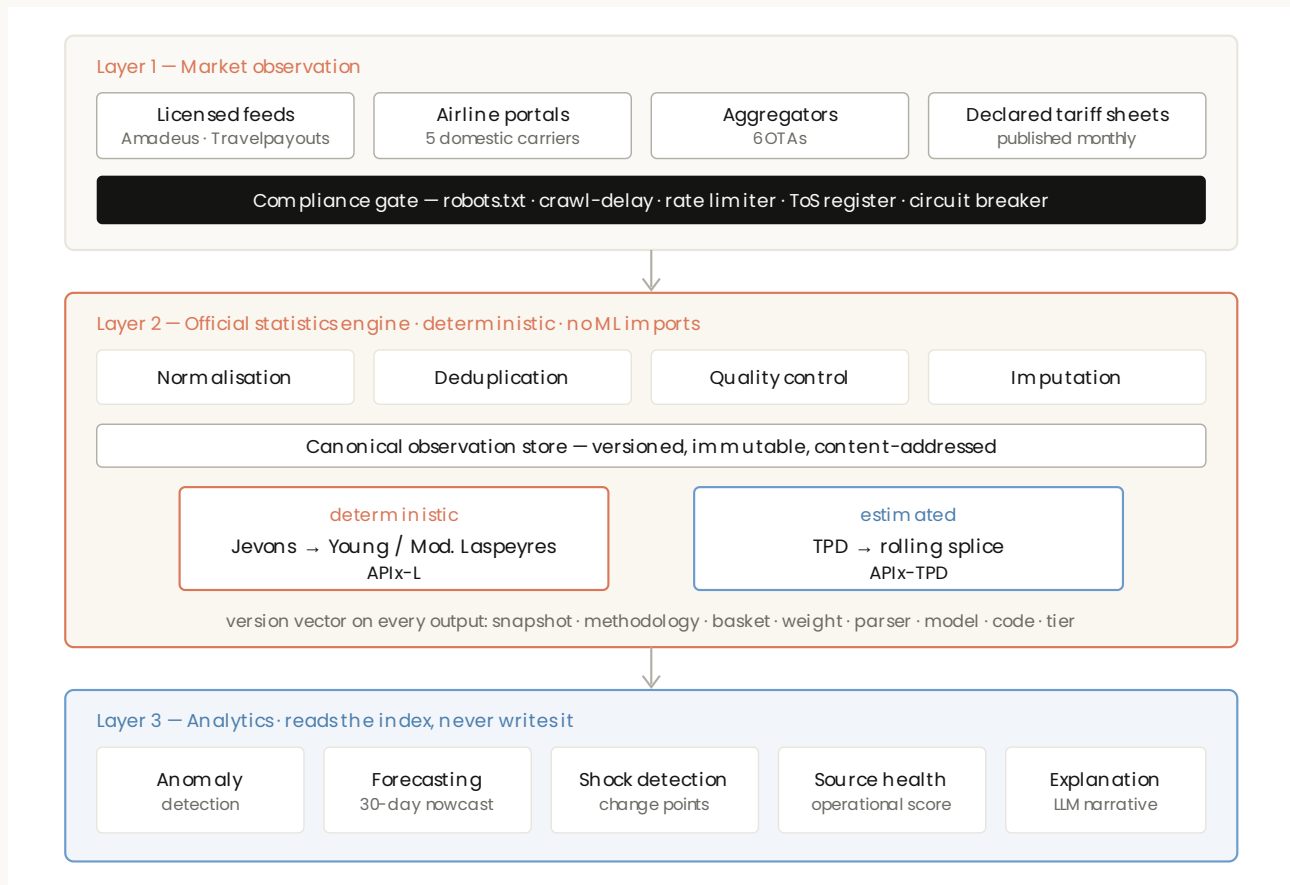
The gap between the two series is the deliverable. It answers: how much of observed airfare movement survives conditioning on product composition, and how much does not?



The two estimators are alternatives, not stages. Jevons output is never fed into the TPD regression; doing so would strip the within-cell variation the hedonic model needs and double-count the aggregation.

System architecture

Three layers with a hard boundary between them. The boundary is not a diagram convention — it is enforced in the repository by a CI import check that fails the build if the statistics package imports anything from analytics or AI.



The engineering rule, stated once and enforced everywhere: statistics calculates, machine learning explains. Remove Layer 3 entirely and the index still publishes.

```
# tests/test_architecture.py — runs in CI, fails the build on violation
FORBIDDEN = {"sklearn", "lightgbm", "xgboost", "torch", "anthropic", "openai"}

def test_statistics_layer_is_deterministic():
    for module in walk_imports("apix/statistics"):
        assert not (module.imports & FORBIDDEN), (
            f"{module.name} imports a stochastic dependency. "
            "The published index must be reproducible from "
            "(snapshot_id, methodology_version, weight_version, code_version) alone."
        )
```

Data acquisition, and the line we do not cross

The problem statement asks the system to handle dynamic CAPTCHAs. APIx treats an access challenge as a stop signal rather than an obstacle. There is no CAPTCHA solver, no fingerprint evasion, and no identity rotation intended to defeat access controls.

Position

A national statistical instrument cannot rest on techniques that violate the terms of the sources it depends on. When a source challenges us, the collector backs off, the event is logged, source confidence is reduced, and a permitted alternate channel takes over. This is a design constraint, not a limitation to apologise for.

Four channels, ranked by evidential strength

Channel	Source	What it gives	Role
A	Licensed APIs	Structured fares with base / tax / fee already decomposed	Supplementary market source · backfill · demo continuity
B	Declared tariff sheets	Route-wise fare categories airlines publish and furnish monthly to DGCA	Reference envelope for observed fares
C	Airline portals	Transactable displayed fare, carrier-direct	Primary observation
D	Aggregator portals	Transactable fare including convenience charges	Channel comparison, consumer-facing price

Source registry

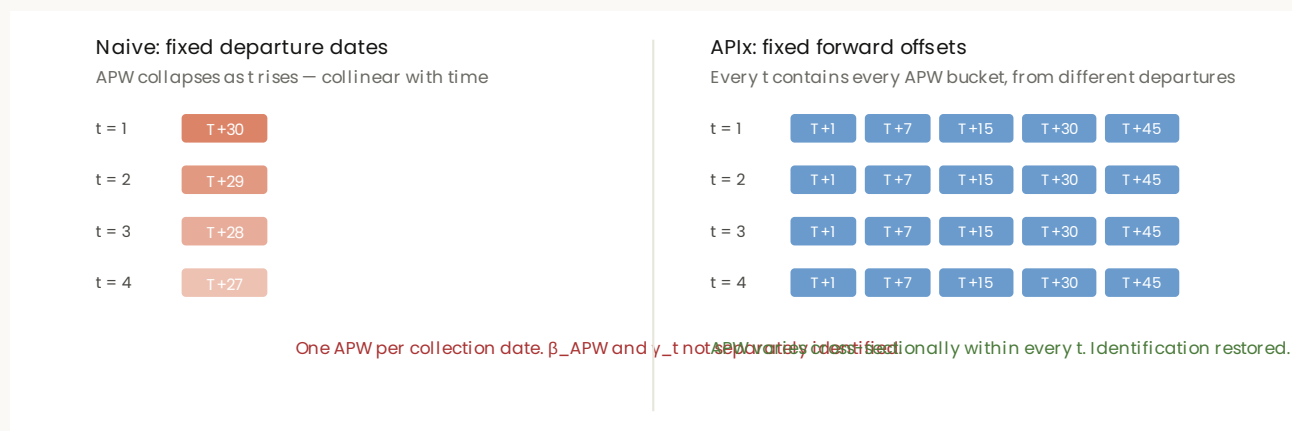
The problem statement names five airlines and six aggregators. The registry records what is committed for the prototype and what is deferred, so no source is silently dropped.

Source	Named in PS	MVP	Note
IndiGo, Air India, Air India Express, Akasa Air, SpiceJet	Yes — 5 carriers	All five	Carrier-direct is the primary observation channel
MakeMyTrip, Goibibo	Yes	One of the two	Same corporate group and shared inventory. Treating them as independent sources would understate correlation and inflate effective N
Yatra, EaseMyTrip, Cleartrip, Ixigo	Yes — 4 aggregators	Two at MVP, four if time permits	Ordered by traffic share; the remainder are deferred, not dropped

Effective sample size is computed on independent sources, not on source count. Two aggregators reselling one inventory feed are one observation, not two.

The sampling design solves an identification problem

If a fixed set of departure dates is observed repeatedly, advance-purchase window falls mechanically as calendar time rises. In a hedonic regression, the APW coefficient and the time effect then become partially collinear — the model returns numbers, but they do not mean what they appear to mean.



The fix lives in the sampling frame, not the estimator. Each collection date samples departures at fixed forward offsets so that advance-purchase window varies within the period as well as across it.

Frame construction

- Routes drawn by probability proportional to size on DGCA city-pair passenger traffic.
- Forced inclusion of the 78-route DGCA-aligned universe. This is a policy-relevant reference frame, not a claim to reproduce TMU's series.
- Strata: metro–metro trunk, metro–tier 2, tier 2–tier 2, and RCS/UDAN routes.
- Advance-purchase buckets $T+1$, $T+3$, $T+7$, $T+15$, $T+30$, $T+45$, $T+60$.

- Departure day type: weekday, weekend, festival calendar.
- Scope decision, stated explicitly: one-way economy fares only in v1. Round-trip fares are not 2× one-way and are deferred rather than assumed.

Planning estimate, pending the seven-day collection spike: roughly 20,000–35,000 quotes per day at full frame. This is small data; the engineering difficulty is comparability, not volume. After the spike this figure is replaced with measured median and p95 daily volume, retrieval success rate, duplicate rate and missingness.

Comparability: making two quotes mean the same thing

Three sources of incomparability sit between a scraped number and a usable price observation. None is exotic. All three will silently corrupt an index if left unhandled, and none appears in the problem statement.

Collection time is part of the observation

Fares move intraday. A quote pulled at 03:00 and a quote pulled at 18:00 are not two draws from the same price; they are two different prices. If one source is collected in the morning and another in the evening, a between-source comparison measures the clock as much as the market.

All collection therefore runs inside a fixed daily window, and the collection timestamp is recorded on every quote. Quotes falling outside the window are flagged and excluded from the index while remaining in the observation store. The window is a published methodology parameter, not an operational detail.

The fare concept must be one thing, not four

Channel C returns a carrier-direct fare. Channel D returns the same seat plus a platform convenience charge. Channel A returns a GDS fare with its own fee structure. Averaging them produces a number that is not the price of anything.

Rule	Treatment
Source is part of the cell key	An elementary cell never mixes channels. Cross-channel differences become a measurable spread, not a hidden contaminant
Canonical fare concept	The index is computed on <i>total payable by an adult passenger for one seat</i> , inclusive of taxes, UDF, PSF and GST, and inclusive of platform charges where the channel imposes them — because that is what the household pays
Decomposition is recorded, not required	Where base/tax/fee split is available it is stored and reported. Where it is not, the total still enters the index. The index is never blocked on a breakdown the site does not render
Channel spread published	The carrier-direct versus aggregator gap is reported as a diagnostic series. It is interesting in its own right and it exposes parser drift early

Fare families do not harmonise themselves

One carrier's saver fare is another's lite is an aggregator's cheapest-available. These labels are marketing artifacts and they change without notice. Mapping them by name is how an index quietly starts comparing a hand-baggage-only fare to a flexible one.

APIx maps to a canonical fare class on **entitlements, not labels** — checked baggage allowance, change and cancellation terms, seat selection. A quote whose entitlements cannot be determined is *excluded*, not guessed into the nearest class, and the exclusion rate is published per source. A rising exclusion rate is the earliest available signal that a site has been redesigned.

Why this section exists

Each of these three is invisible in a demo and fatal in a series. A dashboard built on unharmonised fare families will look correct for weeks and then move six percent because a carrier renamed a fare bundle.

Index methodology

What counts as an item — the question that gates everything else

A matched-model index requires the *same item* observed at t and $t-1$. A specific flight on a specific departure date exists exactly once and never recurs, so the flight cannot be the item. But the cell cannot simply be substituted for it either: if a cell holds one aggregated price per period, a geometric mean across quotes inside it is a **unit value**, not a Jevons index, and unit values are biased whenever the items inside are heterogeneous — the exact condition this document argues holds for airfares.

What APIx defines instead is a **recurring product class**: a bundle of stable observable characteristics that reappears on a fixed cycle. This is an approximation to matched-model comparability, not literal identity — a Monday flight and the same flight number seven days later are different departures under different inventory conditions. The document uses "recurring product class" throughout and never claims the items are the same, because that claim would not survive a statistician's first question.

Whether the recurrence actually exists in the data is measured, not assumed, and the design degrades in declared steps rather than failing.

● Tier 1 — flight-identity match

`carrier × flight_number × day_of_week × APW bucket × fare class
× channel`

A carrier's flight number on a given weekday repeats every seven days, giving price relatives between comparable product classes at weekly frequency — the same frequency at which CPI 2024 collects online prices. Daily APIx is then a weekly-matched index published on a rolling daily basis, not a daily-matched one. That distinction is stated, not blurred.

Used when flight-number identity is stable across the observation window for $\geq 70\%$ of scheduled flights on the route.

Tier 2 — schedule-slot match

carrier × departure_hour_band × day_of_week × APW bucket × fare class × channel

If flight numbers churn but the schedule is stable, the recurring product is the slot rather than the number. Weaker matching, still a matched model.

Used when identity stability falls between 40% and 70%.

Tier 3 — declared unit value

The cell price becomes an explicit unit value, published as such, with within-cell price dispersion reported alongside it as evidence for or against the homogeneity the fallback assumes.

Used below 40% stability. Never used silently: the tier is recorded per cell in the version vector and the Tier-3 share is a headline quality metric.

The tier is a property of each route, measured during the collection spike and re-measured monthly. A route whose tier degrades is flagged; a national index in which the Tier-3 weight share exceeds 25% is published with a prominent quality caveat.

Formulas

Naming a formula is not specifying it. Each aggregation carries a formula, a weight source, a weight reference period, a normalisation rule and a missing-cell rule, all fixed before any index code is written.

The construction is **relative at the elementary level, level at every level above it**. Jevons produces a short-term relative; that relative chains into a cell index level; levels are aggregated. Aggregating relatives directly, or dividing one short-term relative by another, does not produce an index and is the most common way this construction goes wrong.

- 1. Elementary short-term relative (cell c , period t), Jevons
 $M(c,t)$ = items present in BOTH t and $t-7$. Unmatched items enter neither side.

$$J(c,t) = \prod_{i \in M(c,t)} (p[i,t] / p[i,t-7])^{(1 / |M(c,t)|)}$$

- 2. Cell index LEVEL, chained on the cell's own weekly recurrence

$$I(c,t) = I(c,t-7) \cdot J(c,t) \qquad I(c,t_0) = 100$$

If $J(c,t)$ is undefined (no matched items), the parent stratum's relative is carried: $I(c,t) = I(c,t-7) \cdot J(\text{parent}(c),t)$

- 3. Route index, Young / Modified Laspeyres over cell LEVELS

$$I(r,t) = \sum_c v[c|r] \cdot I(c,t) \qquad \sum_c v[c|r] = 1$$

- 4. National index, Young / Modified Laspeyres over route LEVELS

$$APIx(t) = \sum_r w'[r] \cdot I(r,t)$$

Steps 5 to 7 govern the edges — what happens when the population of cells changes underneath the index, which on airline routes it does constantly.

- 5. Renormalisation when cells or routes are suppressed (L = live set)

$$w'[r] = w[r] / \sum_{j \in L} w[j] \qquad \text{for } r \in L$$

- 6. Entry of a new cell: it enters at its PARENT's current level, never at 100. Entering at 100 injects a spurious jump into the aggregate.

$$I(c_{\text{new}}, t) = I(\text{parent}(c_{\text{new}}), t)$$

- 7. Annual re-referencing, same discipline CPI uses across base revisions

$$\begin{aligned} LF &= \text{mean level of new reference year} / \text{old} \\ I_{\text{linked}}(t) &= I_{\text{old}}(t) \cdot LF \end{aligned}$$

A consequence of weekly matching that must be stated, not discovered

Because the matched item includes day of week, each cell advances on its own seven-day chain. The daily series is therefore an aggregate over **seven interleaved weekday chains**, each contributing its most recent level. This is why APIx is published daily but described as weekly-matched, and it is why a Monday and a Tuesday observation are never differenced against each other.

APIx-L chains too, and the document says so

This dossier argues that daily chaining of naive averages accumulates composition effects. APIx-L is itself a chained index, and intellectual honesty requires saying that it is therefore *not* drift-free by construction. What differs is the exposure. Naive chaining links unmatched aggregates, so every change in the quote mix enters the link. APIx-L links matched items only, at weekly frequency, within cells that already hold carrier, fare class, channel and advance-purchase window fixed. The residual drift is bounded by matching quality, and it is measured rather than asserted: scenario A of the controlled

experiment reports the terminal value for the naive index **and** for APIx-L, not only for the naive one.

Anticipate the question directly. "Your headline index chains as well — why is chaining only a problem when they do it?" The answer is that chaining is not the defect; chaining *unmatched* aggregates is. Having the number for APIx-L's own residual drift, on a slide, is what makes that answer credible instead of rhetorical.

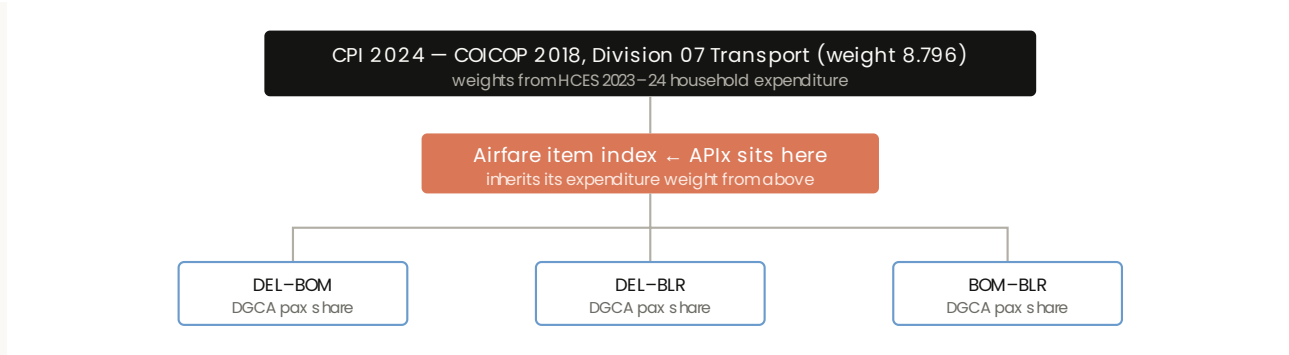
Weight	Source	Reference period	Status
$w[r]$ route	DGCA city-pair passenger traffic	Prior calendar year, lagged and held fixed	Pending verification of public city-pair granularity
$w[c]$ APW	Booking lead-time distribution	Fixed for the index year	No public source identified — see below
Above APIx	HCES 2023–24 household expenditure	CPI 2024 weight reference	Not ours to set; inherited

Two weight sources are not yet secured, and the document says so

Route weights. DGCA publishes monthly traffic statistics, but city-pair-level passenger volumes at the granularity this design assumes may not be openly downloadable. If they are not, the fallback is scheduled seat capacity by city pair, derived from published airline schedules — a proxy for passenger volume with a stated and testable bias. This is a gate item.

Advance-purchase weights. A booking-curve weighting requires a lead-time distribution that no public Indian source is known to publish. Inventing one would be worse than not having one. **v1 therefore uses equal weights across APW buckets as a declared, documented choice**, accompanied by a sensitivity analysis showing how far the index moves under plausible alternative curves. Booking-curve weighting is claimed only once a data source for it exists.

Renormalisation on suppression is the subtle one: dropping a route without renormalising silently reweights everything else. The weight vector is asserted to sum to one in a unit test that runs on every publication.



DGCA passenger shares weight routes *within* APIx. Household expenditure weights sit above APIx and are not ours to set. Conflating the two invites the question of why aviation traffic data is weighting a consumption index.

Quality adjustment — Time Product Dummy

Estimated on the quote-level panel over a rolling window, with time dummy coefficients exponentiated to give the quality-adjusted index, and windows joined by mean splicing so published values are not revised.

$$\ln P[i,t] = \alpha + \sum_t \gamma[t] \cdot D[t] + \sum_k \beta[k] \cdot X[i,k] + \varepsilon[i,t]$$

$$\text{index}[t] = \exp(\gamma[t])$$

Characteristics included

Route, carrier, stops, departure-hour band, flight duration, day-of-week type, advance-purchase bucket, baggage allowance, refundability, fare family, channel.

Deliberately excluded

Seats remaining, availability signals, promotion flags, inventory-depth proxies — variables plausibly jointly determined with demand and fare formation. Conditioning on them would absorb the very movement being measured.

Estimation thresholds — numeric, fixed before implementation

These are not left to whoever writes the estimator. A window failing any threshold produces no published TPD value for that period; the failure and its reason are recorded.

Guard	Threshold	Action on breach
window_length	21 days default; swept over 15 / 21 / 25 / 31	Spread across windows published, not hidden
min_quotes_window	1,500	Window rejected
min_quotes_per_level	30 per characteristic level	Level pooled into an "other" category; pooling recorded
min_quotes_per_route	60 per window	Route excluded from that window's estimation
condition_number	reject above 30	Window rejected as near-singular
max_missing_characteristic	10% of quotes	Characteristic dropped from the specification; drop recorded in version vector
min_periods_overlap	7 days for splicing	Splice deferred; previous window extended
standard errors	clustered by route	—

Missing data and sold-out flights

A sold-out flight is not a missing price. It is a disappeared item, and treating it as an ordinary missing value can bias the index downward where low availability is systematically associated with higher prices. That association is plausible but not assumed: it is tested by comparing realised prices in cells shortly before and after sell-out, and the test result is reported.

Imputation is deliberately conservative and deterministic. A missing cell inherits the price relative of its parent stratum where one is available; where none is, the cell is suppressed and the coverage loss is published. Nothing more sophisticated ships in v1. Model-based completion — temporal matrix factorisation and similar — is a fitted, stochastic method and belongs in the analytics layer as a candidate improvement evaluated against the deterministic rule, not embedded in the index path where it would quietly break the reproducibility guarantee.

Suppression rule	Threshold
<code>min_matched_items_per_cell</code>	3 – below this the cell publishes no relative
<code>max_cell_imputation</code>	40% – above this the cell is suppressed rather than imputed
<code>min_route_coverage</code>	60% of expected cells – below this the route is suppressed and weights renormalised
<code>max_suppressed_weight</code>	15% of total weight – above this APIx publishes with a quality caveat
<code>max_tier3_weight</code>	25% of total weight – above this the unit-value caveat is shown on the headline

Uncertainty

Elementary cells are resampled with replacement and the *entire index path* is recomputed for each draw – chain included, so the serial dependence the chain creates is carried into the interval rather than ignored. Percentile intervals are reported on both levels and growth rates. Bootstrapping single days and reporting the result as an interval on a growth rate would be wrong, and is a common error.

Reproducibility of stochastic procedures

The bootstrap and any numerical routine with stochastic behaviour run from a seed derived deterministically from the version vector, recorded with the output. Two runs of the same publication reproduce the same interval to the last digit. A confidence interval that moves between runs of identical inputs is a defect, not sampling variation.

Compute budget

One thousand draws, each recomputing the full path and refitting TPD over the rolling window, is not free. The cost is measured on synthetic data of realistic size during the verification gate, before the interval is promised on screen. If the run exceeds the publication window, the draw count is reduced and the resulting interval width penalty is reported – not the interval quietly dropped.

The controlled experiment

The central claim — that naive aggregation accumulates composition effects while the quality-adjusted estimator does not — is demonstrated under conditions where the true answer is known by construction, not extracted from a conveniently chosen window of real data.

Design

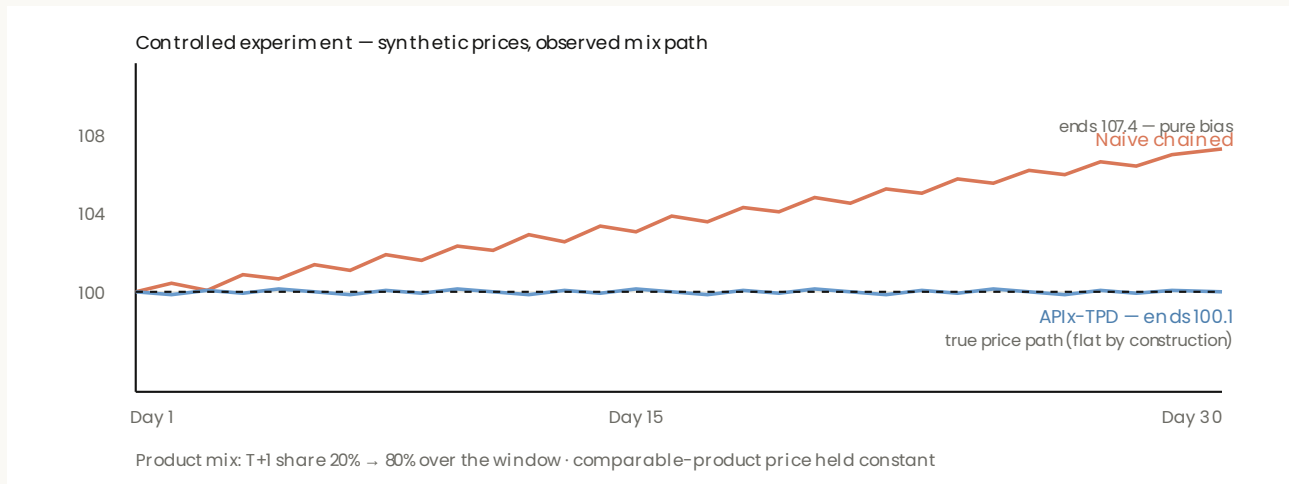
Hold the comparable-product price process fixed by construction. Let the product mix evolve according to the mix path actually observed in the collected data. Run all three estimators on the resulting synthetic panel. The true index is known by design, so any departure from it is measured bias.

One scenario is not an experiment. Six are run, and one of them is designed so the method can fail.

True price path	Perturbation	What it tests
A Flat	Advance-purchase mix shifts	The core drift claim
B +5% over window	Advance-purchase mix shifts	Real movement is not suppressed along with composition
C Flat	Carrier mix shifts	Generalisation beyond the APW dimension
D Flat	Cells go missing at observed rates	Imputation rule under attrition
E Flat	Contaminating outliers injected	Robustness of the flagging rule
F Flat	Model misspecified: an unobserved price-driving characteristic, a nonlinear APW effect the bucket dummies cannot capture, heteroskedastic noise	Whether the method survives being wrong

Why scenario F is the one that matters

In scenarios A through E the data-generating process matches the estimator's specification, so TPD recovers the truth by construction. That demonstrates the arithmetic is correct; it does not demonstrate the method is sound. Scenario F breaks the match deliberately. The result to report is graceful degradation with a stated residual bias — TPD ending at 101.4 against a naive 107 is a far stronger claim than TPD ending at exactly 100.0, because the first is a measurement and the second is a tautology.



Scenario A, indicative shape. Final numbers come from the experiment run on collected mix data, reported for all six scenarios with residual bias stated. The figure is labelled as a controlled experiment on the chart itself, not in a footnote — a clearly declared synthetic demonstration is more credible than a real-data chart that could have been chosen for its shape.

Short-window validation, and what it can honestly support

This is short-window validation, not a back-test, and the document uses that name throughout. A back-test implies a history long enough to test against. Two constraints make that unavailable, and both are stated up front rather than discovered under questioning.

The base period does not overlap

Published CPI is 2024=100 with base prices from calendar 2024. APIx observations begin in 2026. No comparable *level* can be produced. Comparison is on month-on-month and week-on-week growth rates only.

Thirty days is one observation

The problem statement asks for 30 days against monthly data. That is a single monthly point; sixty days gives two. A correlation on $n = 2$ is not evidence. We report direction agreement and magnitude gap with n stated.

Validation hierarchy

Level	Evidence	Strength
1	Controlled experiment, six scenarios including misspecification	Strongest — truth known by construction
2	Weekly growth comparison across the three internal estimators	Strong — many observations, internal consistency
3	Monthly growth against the published CPI airfare item	Weak at $n = 2$; direction agreement and magnitude gap only, with n stated
4	Longer historical validation	Not available in this cycle; named as the next milestone rather than implied

Level 1 carries the argument. Level 3 is reported because the problem statement asks for it, at the strength it can actually bear.

The benchmark mapping is configuration, not code

The exact CPI airfare item label, its item code, its published frequency and its historical range are open verification items. Nothing about them is hard-coded. The benchmark module reads a configuration file, and if the item turns out to be published only at class or sub-class level, the comparison degrades to that level by changing one line rather than by rewriting the validation layer.

Estimator	Frequency	Quality adjusted	Role
Naive daily average	Daily	No	Baseline — shows the problem
APIx-L	Daily, weekly-matched	No	CPI-aligned headline
APIx-TPD	Daily	Yes, under a stated model	Robustness and decomposition
Published CPI airfare item	Monthly	Official methodology	External reference

Alongside these, an ablation suite quantifies how much each design decision moves the index: with and without sold-out imputation, with and without outlier flagging, equal-weighted versus alternative advance-purchase curves, TPD window length across the sensitivity sweep, matched-item tier forced down a level, and a five-day simulated source outage to confirm the index stays inside its interval.

Publication and revision policy

A series without a release calendar is not a statistical product. Anyone evaluating APIx for adoption will ask when the number appears, when it stops changing, and what causes it to change. The answers are fixed in advance.

Stage	Timing	Status
Collection window	Fixed daily window, published as a methodology parameter	Quotes outside it are flagged and excluded
Data cutoff	Close of the collection window	Late arrivals are stored, not included
Provisional publication	Same day, within hours of cutoff	Provisional — carries its own quality triad
Final	T+3 days, after the late-arrival window closes	Final — not revised except under the rules below
Weekly and monthly aggregates	Computed from final daily values only	Never built on provisional inputs

What triggers a revision, and what does not

Revises the series

- A parser defect that produced wrong values, once corrected
- A source correcting data it previously served
- Annual weight update, applied forward from a stated date

Does not revise the series

- A methodology change. That creates a new `methodology_version` and a new series published in parallel, with a linking factor — the same discipline CPI itself uses across base revisions
- Late-arriving quotes after T+3
- A better model becoming available

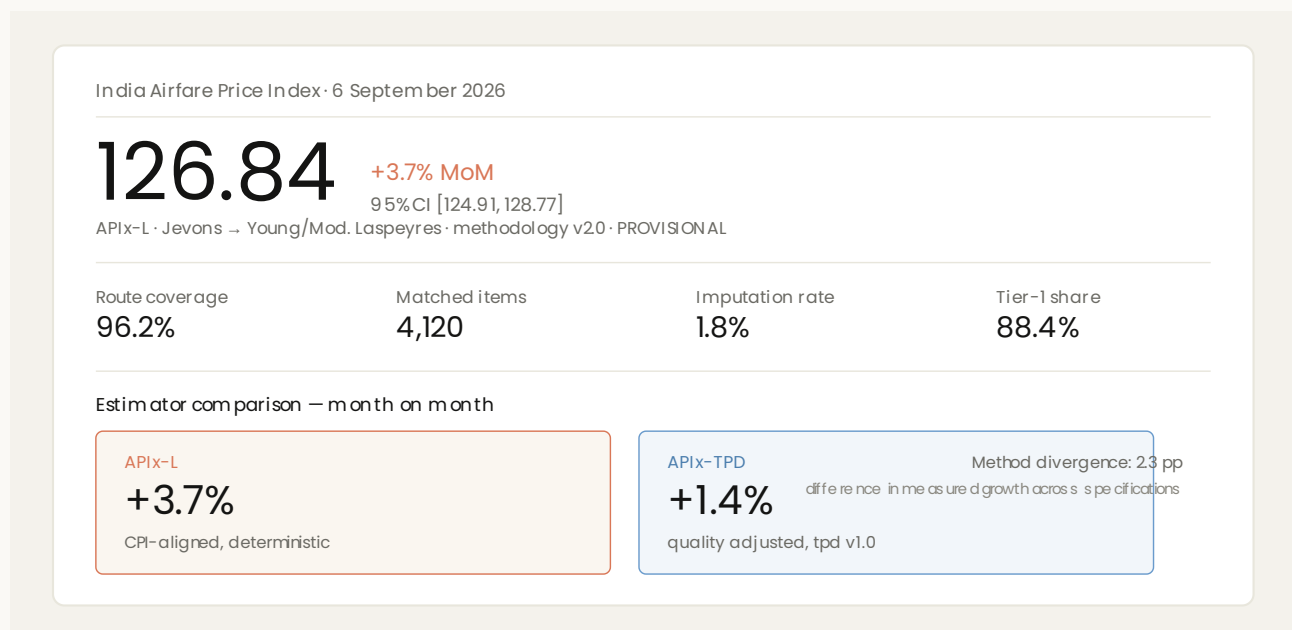
Published values are never overwritten in place. Every revision creates a new vintage; the prior vintage remains retrievable through the API. A user who cited APIx last month can still reproduce exactly what they cited.

Product surface

Five screens. Each answers one question the primary user would actually ask, in the order they would ask it.

"Effective N" is not displayed, because it is not one number

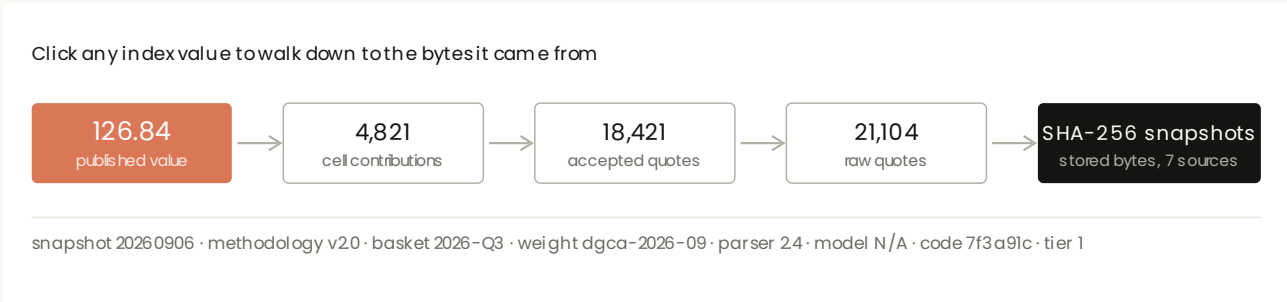
A single effective-sample-size scalar invites a question — Kish? cluster-adjusted? independent product count? — with no good answer unless one is chosen and defended. The headline therefore shows three counts that each mean exactly one thing: **matched items** contributing relatives, **unique offers** observed, and **independent source groups** after collapsing shared-inventory platforms. Operational statistics — sources live, challenge events, parser versions — sit one click deeper, on the provenance screen, so the headline stays readable to a policy user.



Screen one, illustrative values. Growth rates are compared, never levels — the two estimators accumulate along different paths, so a level gap between them is an artifact of divergence since the base period rather than a monthly composition effect.

Screen	Question it answers	Contents
National index	What is the number?	APIx-L, interval, quality triad, provisional/final status, method comparison
Methodology switchboard	How was it computed?	Formula, variables, weights, thresholds, matched-item tier, splicing, outlier and missing-data rules, version — inspectable without reading code
Method comparison	Is the movement real?	Naive, APIx-L, APIx-TPD, the six-scenario controlled experiment
Route intelligence	What moved it?	Contribution waterfall, lead-time elasticity curves, sector heatmap, observed fare against declared tariff envelope, channel spread
Provenance and compliance	Can it be trusted?	Lineage from index value to raw snapshot hash, version vector, vintage history, source register, circuit-breaker state

Provenance is a first-class product feature



Every record additionally carries a `source_type` of `LIVE_SCRAPE`, `AUTHORIZED_FEED`, `PUBLIC_DATASET`, `BACKFILLED` or `SYNTHETIC`, rendered distinctly in the interface. These are never blended silently.

Technology stack

Chosen for what ships, not for what impresses on a slide. Everything below is either load-bearing or cut. Components deferred to a scaling roadmap are listed separately and are not built.

Collection

Python 3.12
Playwright (async)
Scrapy
httpx
Amadeus Self-Service
Travelpayouts

Processing

Polars
PyArrow
NumPy
Pydantic v2
Pandera contracts

Analytics

LightGBM
Isolation Forest
ruptures

Orchestration

cron + Python entrypoint
Docker Compose

Interface

Next.js + TypeScript
Tailwind
ECharts

Explanation

Claude API
strictly Layer 3

Compliance

robots.txt parser
pyrate-limiter
tenacity
registry.yaml

Statistics

SciPy
statsmodels
custom Jevons
custom Young/ML
custom TPD + splice
cell bootstrap

Storage

PostgreSQL 16
MinIO / S3
Parquet
Redis

Serving

FastAPI
SQLAlchemy
SDMX-JSON output

Quality

pytest + Hypothesis
VCR.py cassettes
ruff · mypy
GitHub Actions

Deferred

Airflow
ClickHouse
Kafka / Redpanda
TimescaleDB
Kubernetes
OpenTelemetry

Removal test

Delete the Claude API integration and the entire analytics layer. Collection, cleaning, deduplication, Jevons, Young/Modified Laspeyres, TPD and index publication must all continue to work. If they do not, the layer boundary has been violated somewhere and the CI import test will say where.

What "tested" means in the statistics layer

Line coverage is the wrong target. An index engine can reach full coverage while computing the wrong number. The layer is held to invariants that must hold for any correct implementation, checked as property tests on generated inputs plus golden values computed by hand:

- Identical prices in both periods produce an index of exactly 100.
- Scaling every price by k scales the index by k ; the price relatives are unchanged.
- The index equals 100 in the base period by construction.
- Weight vectors sum to one, before and after any suppression and renormalisation.
- Reordering observations within a period does not change any output.
- Missing cells follow the declared rule and nothing else; a cell is never silently dropped.
- Every threshold in section 08 has a test that crosses it in both directions.
- Recomputing from the same version vector reproduces the published value bit for bit.

The one artifact that must exist before any statistics code

`docs/methodology/apix_formula_spec_v1.md` is frozen before the index engine is written. Not prose about the methodology – the methodology, symbol by symbol. Unit of observation, matched-item definition and tier ladder, Jevons relative, cell level and chain, Young / Modified Laspeyres at route and national level, weight normalisation, new-cell entry, missing cells, suppression, annual linking, TPD specification, splice, bootstrap, versioning. Each with its symbol, definition, reference period, weight source, normalisation rule and edge case.

The purpose is narrow and important: two developers reading the same statistical prose will implement two different indices. A frozen symbol table removes the interpretation.


```
apix/
├─ schemas/           # canonical contracts – written before any collector
├─ ingestion/         # collectors, compliance gate, snapshot store
├─ statistics/        # deterministic – no ML imports, invariant-tested
│   ├─ elementary/    #   jevons.py · matching.py (tier ladder)
│   ├─ aggregation/   #   young_laspeyres.py · weights.py
│   ├─ tpd/           #   specification.py · estimator.py · splice.py
│   ├─ uncertainty/    #   bootstrap.py
│   └─ index/         #   apix.py · publication.py · vintages.py
├─ analytics/         # anomaly · forecasting · shock · decomposition
├─ ai/                # explanation, parser diagnosis, schema mapping
├─ api/               # FastAPI, SDMX serialisers
├─ dashboard/         # Next.js
├─ experiments/       # six controlled scenarios, ablations
├─ docs/              # METHODOLOGY.md → methodology note PDF
└─ tests/             # incl. test_architecture.py, property tests
```

Build sequence and gates

The largest unknown is not the software architecture. It is whether the observations that can actually be collected support the statistical unit the design assumes. Four gates close that question before the collector fleet gets built, and each gate has a numeric pass criterion and a named fallback. A gate that fails does not stop the project; it selects a different branch that was designed in advance.

Gate	Pass criterion	If it fails
1 · Primary sources	Every attribution to MoSPI evidenced by a document on file; airfare item, code, frequency and weight located; DGCA route universe documented	The unevidenced claim is removed from the deck, not softened. Benchmark degrades to class or sub-class level via configuration
2 · Access	Compliance register populated with dated, real entries for every committed source; licensed feed confirmed to cover Indian domestic sectors with required fields	Source moves to deferred in the registry. MVP proceeds on the remaining permitted set
3 · Statistical proof	Drift appears in scenario A; degradation is measured and bounded in scenario F; bootstrap runtime fits the publication window at the intended draw count	Draw count reduced with interval-width penalty reported. If scenario F degrades badly, TPD is demoted from headline comparison to a diagnostic
4 · Live spike	Retrieval success $\geq 85\%$ per source-day; flight identity stable for $\geq 70\%$ of scheduled flights; fare entitlements determinable for $\geq 80\%$ of quotes; Tier-1 match coverage — matched items surviving fare class, APW, channel and entitlement conditioning, over expected recurring items — above a floor set from the spike itself, not invented now	Matched-item tier drops per the ladder in section 08. Below 40% stability the index publishes as declared unit values with dispersion reported

Gate 4 is the one that can reshape the design. If flight identity is not stable across days, the Tier 1 matched item does not exist in the data and the elementary index runs at Tier 2 or Tier 3. Better to learn that in hour six than hour twenty-six — and better to have already written the ladder than to improvise it at hour twenty-six.

Identity stability and match coverage are different quantities and both are needed. Flight numbers can be perfectly stable while the usable matched set collapses after conditioning on fare class, advance-purchase bucket, channel and determinable entitlements. Stability says the product recurs; coverage says enough of it survives to

compute a relative from. The second is the statistic-driving metric, and its floor is measured during the spike rather than asserted in this document.

Sequence

Phase	Work	Exit condition
Gate work	Schemas and contracts first. Then statistical core on synthetic data — Jevons, Young/Modified Laspeyres, invariants, six scenarios, bootstrap timing. Then licensed-feed and access verification. Then the seven-day collection spike.	All four gates resolved to a pass or a named branch. Data contract frozen.
Hours 0–8	Vertical slice: one collector, Bronze → Silver → Gold, matched-item tier assignment, Jevons, Young/Modified Laspeyres, one API route, one chart.	One real index number, from one real collection run, rendered in a browser.
Hours 8–20	Breadth: remaining collectors, compliance gate, outlier and imputation modules, TPD estimator with thresholds, bootstrap intervals, validation notebook.	Both estimators publish; divergence chart populated from real data.
Hours 20–32	Surface: five screens, SDMX endpoint, methodology switchboard, provenance walk, vintage history, ablations, documentation.	Every claim in the pitch has a screen behind it.
Hours 32–36	Freeze. Backup recording. Five full rehearsals. Q&A drill. Placeholder sweep.	No code changes. Demo runs identically five times. No illustrative number survives in any external artifact.

Ownership

Seat	Owns
Statistician	Index engine, matched-item ladder, methodology note, six scenarios, validation. The critical path for every claim that wins — staffed with a second pair of hands from the ML seat during gate work
ML engineer	Controlled experiments alongside the statistician, then outliers, imputation, forecasting, parser diagnosis
Data engineering ×2	Collectors and compliance gate; contracts, medallion layers, snapshot store
Full stack	API, SDMX, five screens
Lead	Integration, gate tracking, deck, demo script, Q&A preparation

Risk register

Risk	Likelihood	Response
Flight identity unstable — Tier 1 matched item does not exist	Medium	Tier ladder in section 08; index degrades to schedule-slot or declared unit value with dispersion reported
DGCA city-pair passenger data not publicly available at required granularity	Medium	Scheduled seat capacity by city pair as weight proxy, with stated bias; gate 1 item
No source exists for booking lead-time distribution	High	Equal APW weights declared in v1 with sensitivity band; booking-curve weighting claimed only when sourced
Collectors blocked during the demo	High	Licensed channel plus pre-seeded database; recorded fallback; imputation demonstrated live
Airfare item index not published at usable granularity	Medium	Configuration-driven benchmark degrades to class or sub-class level
Fare families unmappable across sources	Medium	Entitlement-based mapping; unmappable quotes excluded with published exclusion rate
Site redesign breaks a parser	High	Contract tests, parser version pinning, exclusion-rate alarm, LLM-assisted diagnosis with human approval
Residual chain drift in APIx-L itself challenged in Q&A	Medium	Scenario A reports the terminal value for APIx-L alongside the naive index; the number is known before the pitch, not defended rhetorically
Two developers implement the aggregation differently	Medium	Frozen formula specification with symbol table; golden values computed by hand before either implementation
Bootstrap exceeds publication window	Medium	Draw count reduced; interval-width penalty reported rather than the interval dropped
Legality challenged in Q&A	High	Compliance register on screen; conservative language; no bypass techniques to defend
Team overbuilds and nothing integrates	Medium	Vertical slice by hour 8; hard freeze at hour 32; deferred stack stays deferred

Open verification items

None of the following can be settled by further design discussion. Each requires a primary document or a live collection run, and each gates a claim that appears in this dossier.

- ☐ Read the Expert Group Report on Comprehensive Updation of the CPI. Confirm or drop any attribution of advance-purchase stratification to MoSPI's existing methodology.
- ☐ Pull the airfare item index from the eSankhyiki macro-indicators module. Confirm it is published at item level with usable history; record its label, code and frequency in benchmark configuration.
- ☐ Download the CPI 2024 item weights and locate the exact airfare item and its weight.
- ☐ Establish whether DGCA city-pair passenger volumes are publicly available at the granularity the route weights assume. If not, build and document the scheduled-capacity proxy.
- ☐ Search for any published Indian domestic booking lead-time distribution. Absent one, freeze equal APW weights and build the sensitivity band.
- ☐ Confirm licensed feed coverage of Indian domestic sectors, field availability and current terms — as a supplementary and backfill source, never a silent substitute for the observation being claimed.
- ☐ Run the collection spike: one route, two carriers, one aggregator, five advance-purchase buckets, seven days. Measure retrieval success, base/tax decomposition availability, flight identity stability, fare entitlement determinability, sold-out behaviour, duplicate rate, page latency and access-challenge frequency. The matched-item tier is set by the identity-stability number.
- ☐ From the spike, compute Tier-1 match coverage after full conditioning and set its floor. Identity stability alone does not establish that the elementary index is computable.
- ☐ Measure bootstrap runtime on synthetic data of realistic size before the confidence interval is promised on any screen.
- ☐ Freeze `docs/methodology/apix_formula_spec_v1.md` before the first line of statistics code.
- ☐ Fix the collection window and record it as a published methodology parameter.

APIx — pre-product engineering dossier v2.1. Figures marked illustrative or indicative are placeholders pending the verification items in section 16 and must be replaced with measured values before any external presentation. Thresholds in sections 08 and 14 are defaults fixed before implementation; changing one changes the methodology version.

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